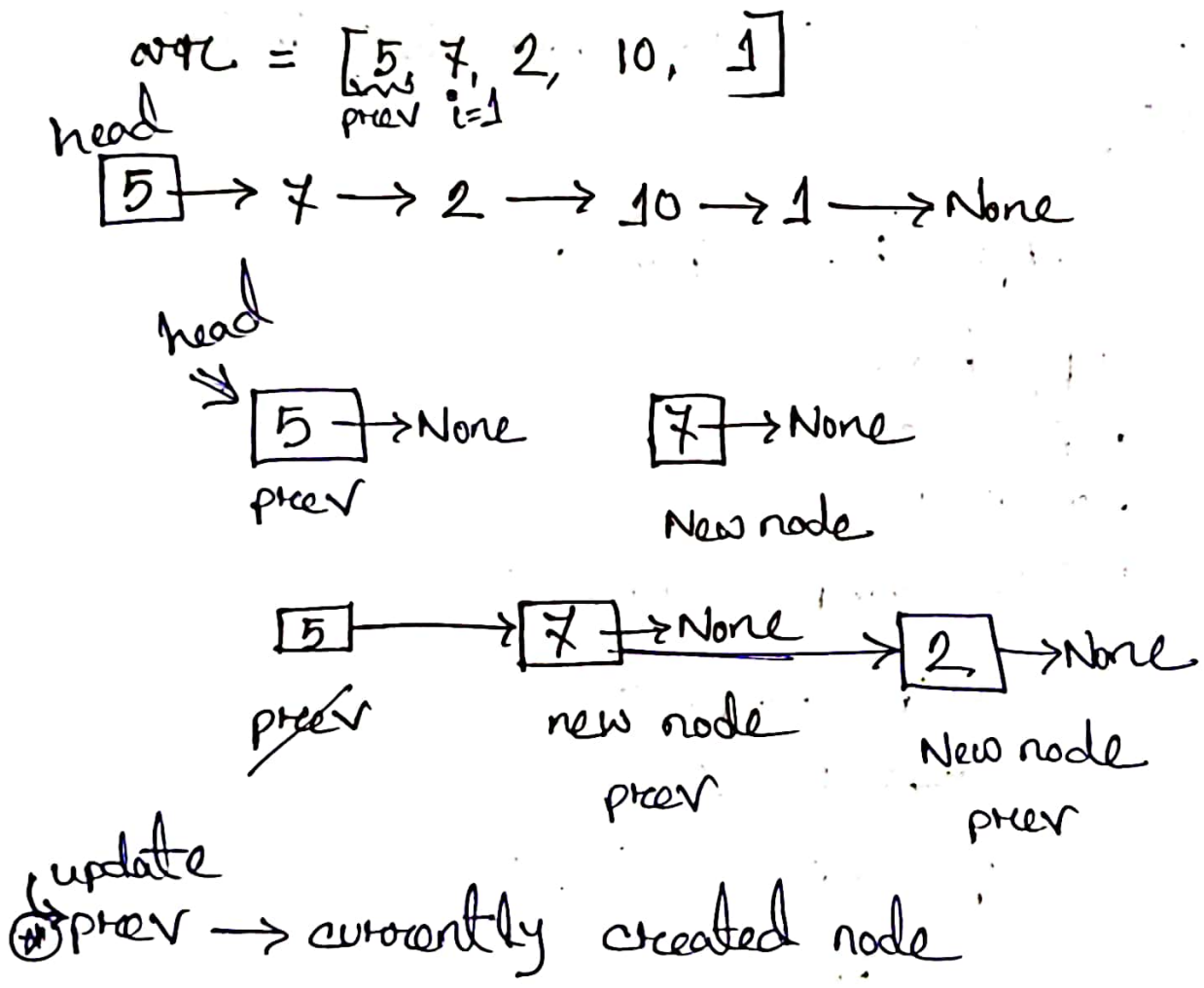
14-06-23

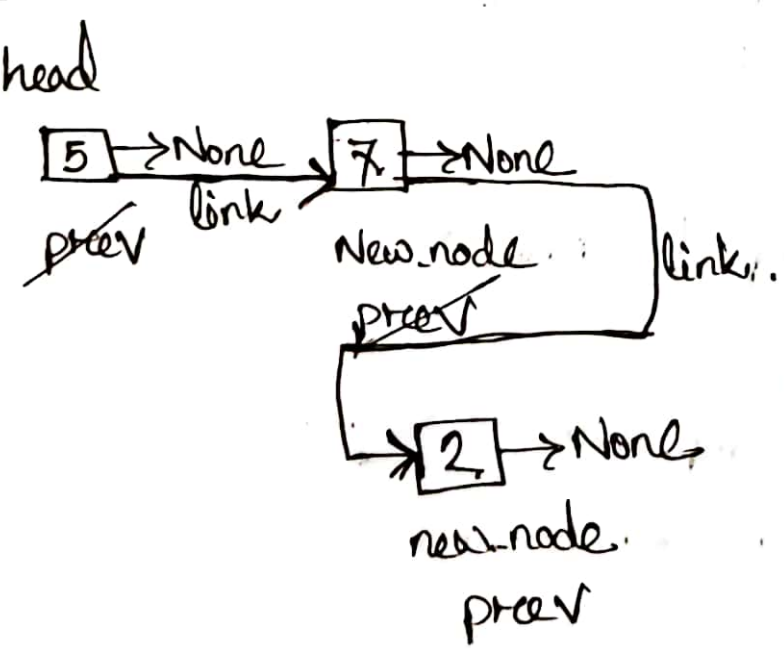
Linked List

Creation of a linked list from an array:



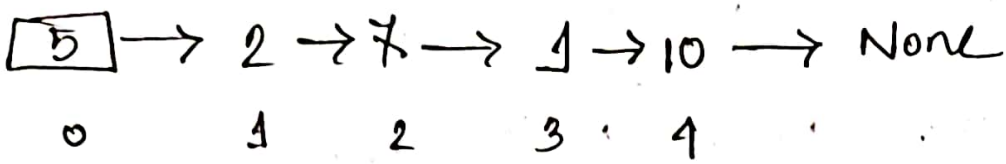
```
def creation(arr):
    head = Node(arr[0], None)
    prev = head  #head node created#
    for i in range(1, len(arr)):
        new_node = Node(arr[i], None)
        prev.next = newnode
        prev = new_node
```

Simulation



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(M) head



nodeAt (head, 3) ⇒ 1 (# value of idx no 3)

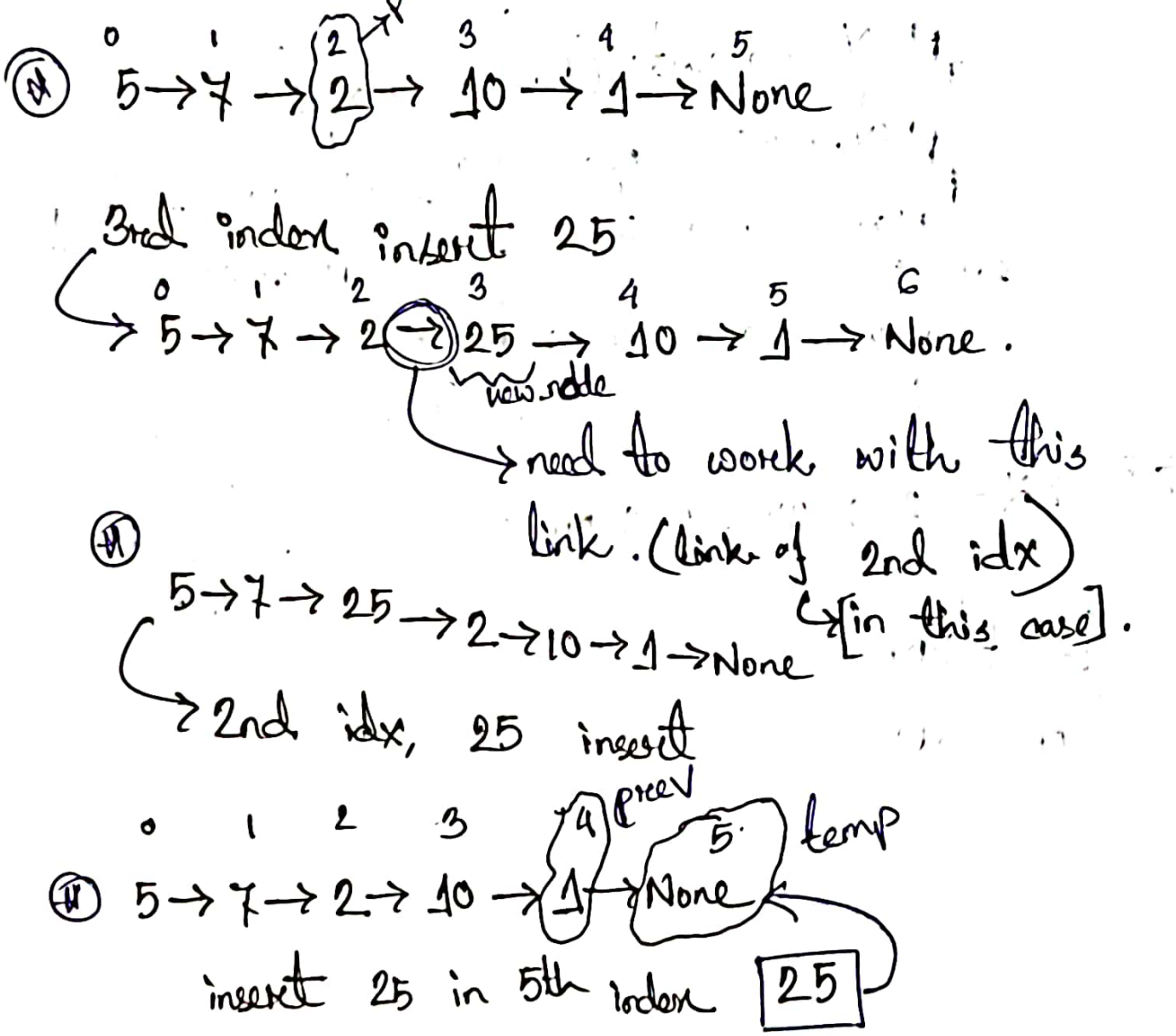
~~node At (head, 7)~~

index of (head, 7) ⇒ 2 (# idx num for value 7)

(*) Returns the index of the given value in linked list:

```
def index_of (head, value):
    temp = head
    count = 0
    while temp != None:
        ~~temp~~.
        if temp.elem == value:
            return count
        temp = temp.next
        count += 1
```

⑯ Insert a value in a specific index in linked list:



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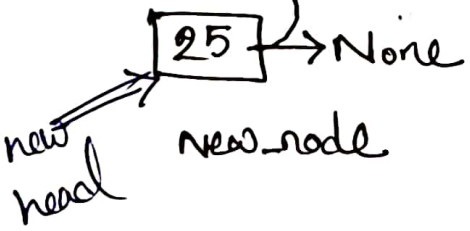
```
def nodeAt ( ):
def insert (head, idx, value):
    new_node = Node (value, None)
    if idx != 0:
        prev = nodeAt (head, idx-1)
        temp = prev.next
        prev.next = new_node
        new_node.next = temp
    else:
        new_node.next = head
        head = new_node.   (# in order to insert value at first idx)
```

Alternative

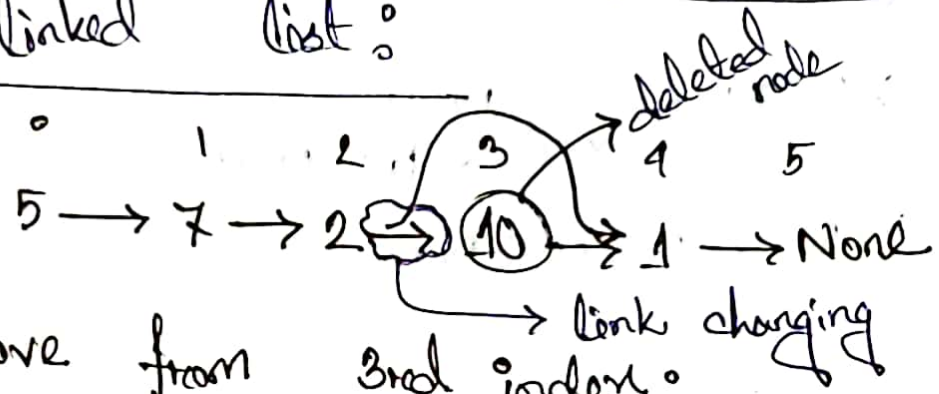
```
new_node.next = prev.next
prev.next = new_node.
```

Now

5 → 7 → 2 → 10 → 1 → None



(#) Remove from a specific index in linked list:



remove from 3rd index:

↳ 5 → 7 → 2 → 1 → None

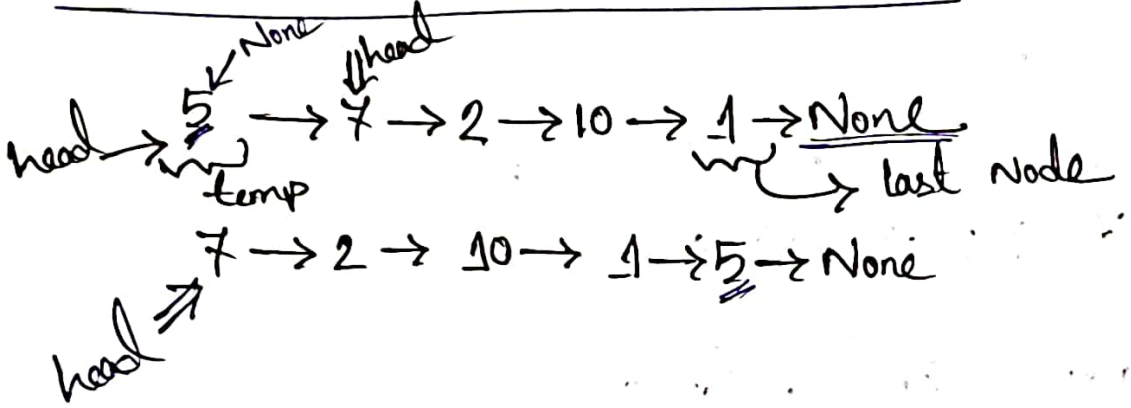
Code: def nodeAT():

```
def remove (head, idx):
    if idx != 0:
        prev = node.At (head, idx-1)
        deleted_node = prev.next
        prev.next = deleted_node.next
        deleted_node = None
    else:
        head = head.next

    return head
```

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⑧ Left Rotate a Linked List:



i) temp = head

ii) head = head.next

iii) last node identify.

iv) last_node.next = temp

v) temp.next = None

Quiz → Next Wednesday

1) Array
2) 2D Array
3) Linked list.